CredLens: Financial Health & Financing Readiness Assistant

Milestone-1 Report: Project Proposal, Requirements and Feasibility

Group: 36

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1. Final Problem Statement and Motivation

Problem statement. A small-business owner may have financial statements without a clear view of what they say about the business. Before approaching a lender, the owner needs to understand the main financial changes, check the figures behind them, and know which questions still need an answer. Our project focuses on this preparation stage, initially for small manufacturing businesses with existing records. [1]

Core problem. The difficulty is turning financial records into findings that an owner can check and act on, while separating financial concerns from information that is simply missing.

Motivation. We want to make financial figures easy to check, keep calculations reproducible, and connect each finding to a useful next step. The user should receive a plain-language summary and a preparation checklist, rather than an unexplained score.

2. SDG Mapping and Rationale

SDG target	Relevance	Rationale and limit
SDG 9.3 (primary)	Preparation for access to financial services by small-scale enterprises. [4]	Measure understanding and preparation, not actual access to credit or indicator 9.3.2.
SDG 8.3 (secondary)	Indirect support for MSME development through clearer financial preparation. [5]	Do not claim effects on employment, growth or formalisation.

3. Background and Context

CredLens will accept one to three years of balance sheets and profit-and-loss statements. After the user reviews the extracted figures, it will calculate six indicators, explain supported findings, identify missing information, and produce a report to discuss with an accountant or lender. Cash-flow statements are optional.

This is not a loan-eligibility tool. Actual applications require more than financial ratios: HDFC Bank, for example, also lists bank statements, tax information and business documents. Approval remains the lender's decision. [3]

4. Literature and Existing-Solution Review

We reviewed work on financial question answering and public product descriptions. The review supports our design choices; it does not prove that CredLens is the first product of its kind.

4.1 Literature review

Study	What we take from it
Lewis et al., 2020 RAG [6]	Retrieval supplies outside evidence for language generation. We will retrieve report passages, but validate figures and calculations separately.
Chen et al., 2021 FinQA [7]	Financial questions are paired with numerical reasoning programs. This supports making arithmetic explicit and testable rather than leaving it to the LLM.
Zhu et al., 2021 TAT-QA [8]	Questions draw on both tables and narrative. We need to preserve period labels, units and table context alongside the source text.
Islam et al., 2023 FinanceBench [9]	Financial answers are evaluated against supporting evidence. We will test factual support, not just fluent wording. Historical benchmark results do not establish our model's performance.

4.2 Existing solutions

Platform	Published capabilities and implication
OneNDF [10]	Business-health reporting, ratios and improvement recommendations. Financial-readiness advice alone is not new.
ACTICO [11]	AI-assisted extraction and mapping, source traceability and human correction. Reviewable extraction is an established capability.
Moody's [12]	Financial spreading and scoring with human expertise and an audit trail. We are not trying to reproduce an underwriting platform.
Fathom [13]	Financial analysis, reporting, forecasting/scenarios and AI commentary. A dashboard or AI report is not enough to establish a USP.

These are vendor descriptions, not hands-on comparisons. If a feature is not mentioned on a website, we treat it as unknown, not absent.

5. Gap Analysis

Our focus: Evidence to Action. Each finding will show its source, calculation or rule, evidence status, and next step. A dependency map connects missing or disputed values to the analyses they block. This gives the owner a specific task, such as providing interest expense, instead of a generic request to upload more documents.

The gap we will test is whether this workflow handles incomplete information better than a basic financial-document chatbot. We will compare basic RAG, RAG with calculations, and the full system. The contribution is the implemented and evaluated workflow, not a claim that no other platform offers similar features.

First request corrections to inconsistent inputs, then missing values that unlock blocked analyses, then investigation of observed changes. This is a workflow priority, not a credit-risk ranking. A falling margin can prompt a review of expenses; it cannot establish the cause without evidence. Recommendations help users investigate, prepare or improve reporting, and do not guarantee loan approval.

6. Stakeholder Analysis

Stakeholder	Role and need
Business owner / finance manager	Primary user: upload statements, check figures, understand findings and prepare for a financing discussion.
Accountant / finance reviewer	Review figures and the exported report; a separate collaboration portal is not required.
Faculty and student team	Faculty assess the design and evidence. The three-member student team develops, tests and documents the application.

7. User Requirements

Users need to use existing records without bank connections (UR1), correct extracted figures (UR2), understand findings (UR3), identify missing evidence (UR4), obtain a report (UR5), and keep business data private (UR6). A walkthrough with 2-3 users and a finance reviewer is planned; access is not yet confirmed.

7.1 Functional requirements

ID	Required behaviour and acceptance check
FR01	Protect business workspaces. Changing a record ID must not expose another business's files, figures, chat or reports.
FR02-03	Accept supported PDFs with year, unit and basis; extract candidate values with page references. Reject unsupported/oversized files clearly and flag unresolved fields.
FR04	Let users correct or enter values. Save origin, reviewer and revision; distinguish user-provided figures from document-supported ones.
FR05-06	Calculate six indicators and check their dependencies. Missing interest expense blocks coverage, not unrelated supported calculations.
FR07-08	Generate evidence-linked findings and next steps; answer focused questions with citations. Numerical answers use tested calculations, not invented values.
FR09-10	Create a printable preparation report. After corrections, refresh affected results and mark older reports as superseded.
FR11-12	Delete active documents and derived content. Log actor, action and time for key events without recording full financial payloads.

Traceability: UR1 -> FR02; UR2 -> FR03-04; UR3 -> FR05/08; UR4 -> FR06-07; UR5 -> FR09-10; UR6 -> FR01/11-12. Retain these IDs in the test log.

7.2 Non-functional requirements

Category	Requirement and check
Correctness / traceability NFR01/04	All formula tests pass. Record inputs, units, periods and formula versions; check source IDs and numbers. Human review checks whether citations support the claim.
Privacy / access NFR02/03	Test local-only processing after setup and at least 20 negative access cases. Enforce ownership on every API and retrieval path. [17,20]
Usability / reliability NFR05-08	Use text labels, not colour alone. Cap uploads at 20 MB/50 pages and one job; record latency/memory and freeze versions. Show failures explicitly; keep reviewed calculations and a template report available.

8. Technology and Tool Selection

We will build one FastAPI application with separate document, financial-analysis, **RAG/retrieval**, and reporting modules. A single application is easier for this team to test than microservices. React provides the review screen, dashboard, chat and report view.

Tool	Why we will use it
React + Tailwind / Vite	A browser interface with reusable review and dashboard components; no separate mobile app.
Python + FastAPI / Pydantic	One backend language for extraction, arithmetic and AI calls, with typed validation and API documentation. [19]
PostgreSQL + pgvector SQLAlchemy / migrations	Store values, findings and searchable document passages together. Use pgvector for retrieval of relevant, authorised passages for the RAG-based Q&A workflow, without a separate vector database. [16]
pdfplumber	Extract page-level text and tables from digital PDFs. Restrict supported layouts and retain manual correction. [15]
Decimal + Python rules	Run the six formulas, comparability checks and input-dependency rules outside the LLM.
Sentence Transformers + Ollama	Generate local embeddings for document retrieval and use a pretrained local instruction model for evidence-grounded Q&A. MiniLM is a candidate; keep chunks within its documented limit. No training is required. [17,18]
Pytest + Git	Test calculations, APIs and gaps, and record changes. Docker Compose is optional packaging; CI/CD is not required.

Records and execution. Store users/businesses, documents/pages/chunks, financial values, analysis runs/findings, chat, reports and audit events. Version rules and outputs. Process one bounded job at a time with progress, retry and failure states; interrupted jobs must not appear complete.

9. Feasibility Analysis

9.1 Technical feasibility

The design is feasible as a student prototype because it needs no default-labelled lending data, new model training or commercial integrations. The main work is extraction, financial rules and reliable explanations. Start with a manual-input version and one complete company case before adding more PDF layouts.

The project will be developed iteratively by three students, with the workload managed through short development sprints and regular integration and testing. Reuse a student or lab computer; 16 GB RAM is a preference, not a measured minimum. Test a small quantised model early, recording licence, memory and latency, and shorten context before considering new hardware.

9.2 Economic feasibility

The proposed additional cash allowance is INR 0-1,500: up to INR 500 for printing and INR 1,000 contingency. No paid API, hosting or new computer is assumed. Existing equipment, electricity, internet and student time are excluded. These are planning allowances, not price quotations.

9.3 Operational feasibility

The owner and accountant can use one interface/report. Users without statements or with scanned-only records are outside this first version. If target users are unavailable, faculty/student feedback can test usability but cannot establish market demand.

9.4 Legal and ethical feasibility

Business statements may be confidential and may contain personal information. Use fictional data and limited public excerpts first; obtain permission before a private-data pilot. Check applicable DPDP provisions and effective dates, institutional rules, notices, retention, deletion and source/model licences before deployment. Not every company figure is personal data. [22,23]

Avoid unnecessary identifiers, keep processing local, and distinguish active-data deletion from backup erasure. Do not claim compliance certification or authenticated accounts. Never suggest hiding or altering records, infer repayment ability from owner characteristics, or turn missing data into an adverse credit judgement.

Hash passwords, protect uploads, keep secrets out of Git and minimise logs. Require HTTPS before any non-local pilot. Treat PDF text as untrusted data; the LLM must not receive execution privileges. [17,20,21]

10. Scope and Limitations

In scope. The MVP handles one business at a time, one to three completed years, English digital PDFs, INR units and standalone statements. Balance sheets and P&L statements are the main inputs; cash-flow statements and borrowing notes are optional. Manual entry/correction is the fallback. One year supports a snapshot; trends need comparable years.

Outside the MVP: loan approval or eligibility prediction, default scores, lender matching, bank/GST/bureau links, authenticity checks, scanned-PDF OCR, multi-currency and consolidated groups. Cloud deployment, CI/CD, forecasting and scenario simulation remain future scope.

11. Risk Analysis and Mitigation (Planning Estimates)

Risk	Likelihood	Impact	Mitigation / responsible role
R1: Wrong PDF values	High	High	Document lead: source-linked review, manual fallback and checked extraction cases.
R2: Wrong formulas	Medium	High	Finance/backend lead: explicit definitions, unit tests and finance review.
R3: Unsupported AI claims	High	High	AI lead: approved results, checked citations, template fallback and unsupported-question tests.
R4: Data disclosure	Medium	High	Backend lead: ownership checks and protected storage; test access boundaries. [20]
R5: Prompt injection	Medium	High	AI/backend leads: upload limits, no LLM execution rights and adversarial tests. [21]
R6: Limited hardware	Medium	Medium	Integration lead: early runtime trial, smaller model/context and calculation-only fallback.
R7: Scope overload	High	High	Coordinator: weekly gates; retain five feature groups and cut optional work.
R8: Misleading advice	Medium	High	Finance/AI leads: cautious wording, no approval promises and reviewed findings.

Risk	Likelihood	Impact	Mitigation / responsible role
R9: Evaluation bias	Medium	Medium	Evaluation lead: held-out cases, equal evidence across variants and labelled proxy-user results.
R10: Stale data / deletion	Medium	High	Backend lead: version results, supersede reports and test deletion of files and derived data.

12. Proposed Solution and Methodology

12.1 How it works

Upload statements -> Review figures -> Analyse health -> Identify gaps -> Explain next steps -> Generate report. The owner can discuss the report with an accountant or lender, then supply missing information and run the affected analysis again.

Two information paths. Numerical questions use reviewed database values and Python results. Document questions use a retrieval-augmented generation (RAG) workflow, retrieving authorised passages with page/section context. The answer assembler combines them; the LLM explains evidence but cannot run arbitrary code, SQL or financial updates.

12.2 Financial calculations and validation

Python will calculate the following measures from reviewed values. The team will have the definitions checked by a finance reviewer. ACCA notes that ratio definitions and interpretation vary; none of these measures is a universal lending threshold. [14]

Indicator	Project definition	Input rule
Revenue growth	100 x (current revenue - prior revenue) / prior revenue	Two comparable periods; prior revenue must be positive.
Net profit margin	100 x profit after tax / revenue from operations	Use a consistent basis; zero revenue makes the ratio undefined.
Working capital	Current assets - current liabilities	Show an INR amount, not a score.
Current ratio	Current assets / current liabilities	Do not divide by zero or apply a universal acceptable level.
Debt-to-equity	Defined interest-bearing borrowings / total equity	Count current maturities once; exclude and disclose lease liabilities. Flag non-positive equity.
Interest coverage	Reviewed EBIT / interest expense	EBIT is not EBITDA. Missing or zero interest is not an infinite or favourable score.

Each figure keeps its raw value, normalised INR value, unit multiplier, period, reporting basis, source page, review status and revision. Check units, years and assets against liabilities plus equity within a stated rounding tolerance. Passing these checks does not authenticate a statement. Optional operating cash flow can be shown with comparable-year changes; absent cash-flow data means cash generation is not assessed.

12.3 Evidence states and next steps

State	Meaning for the user
Supported	Required inputs are reviewed and comparable. Calculate, showing the source or labelled manual origin.

State	Meaning for the user
Needs review	A required value is unconfirmed or conflicting. Resolve it before publishing the dependent conclusion.
Not assessed	A required value or comparable period is missing. Name the missing input and the analysis it blocks.
Not meaningful	The formula or interpretation is unsuitable, such as a zero denominator or non-positive equity.

12.4 Data and evaluation plan

Create four fictional manufacturing businesses with up to three years of balanced statements and selected notes. Keep the correct values and expected results. Make controlled variants with missing interest, wrong units, mismatched periods, unreviewed fields and conflicting figures; change one factor at a time where possible.

Use selected standalone statement pages from three HUL reporting years to test real layouts. Its official archive is the source, but a large listed company is not representative of all MSMEs. Small FinQA/TAT-QA samples can supplement financial Q&A tests; they are not loan/default labels or a full benchmark evaluation. [7,8,24]

Record document, page, year, unit, expected value, review state, formula result and supported explanation. One member annotates and another checks. Seek finance review for definitions and sample findings. Measure extraction before manual correction, and correction effort separately.

Comparison that tests the USP

Variant	What changes
A - basic RAG	The selected model retrieves evidence and answers without a calculation or dependency engine.
B - RAG + calculations	Adds the six deterministic functions.
C - full CredLens	Adds evidence states, comparability checks, blocked-analysis explanations and targeted next steps.

For reasoning tests, give all variants the same documents and reviewed values, serialised where needed. Keep the model, questions and retrieval settings fixed. Test extraction separately so its manual-review advantage is not credited to the gap engine.

Hold out at least one business or layout before tuning. Freeze 60 questions: 12 each for supported numerical questions, supported document questions, missing inputs, inconsistent inputs/periods, and unsupported causal or lending claims. Save expected answers and acceptable refusals before running tests.

Measures and acceptance checks

Measure	Evidence to collect
Calculations	At least 30 assertions for formulas, missing data, signs, units, zero denominators and comparability; all must pass.
Answers and evidence	Numerical accuracy with rounding tolerance; supported/unsupported claims and citation support, checked by human reviewers.
Missing-data handling	Correctly blocked analyses and useful follow-up requests. Fixed dependency cases must behave as specified.

Measure	Evidence to collect
Useful answer coverage	Correct answers to answerable questions, plus false refusals. Refusing everything is not success.
Privacy and use	All defined access tests pass; offline processing and active-data deletion verified. Record task completion, correction effort, time, latency and memory.

Have two reviewers check a sample and resolve disagreements against the evidence. Where feasible, repeat generation and compare manual review with CredLens on matched tasks. Report sample size, participant type and actual results, including failures. A small study cannot establish commercial safety or improved loan approval.

12.5 Development approach

We will use an agile, iterative development approach, delivering the project in five phases. Work will be organised into short sprints with frequent integration, testing and review of working features. The non-AI financial workflow will be implemented first so that model integration does not hold up the core calculations and evidence checks.

Phase	Work and completion check
Phase 1: Core workflow	Confirm the six financial formulas and input rules. Implement manual data entry, one fictional company case, the financial calculations and basic results view. Verify calculations and missing-input behaviour with tests.
Phase 2: Data, security and evidence rules	Implement protected business workspaces, database storage, reviewed-value states, source/origin tracking and dependency/gap rules. Demonstrate that missing or conflicting inputs block only the affected analyses and that cross-business access is denied.
Phase 3: PDF upload and review	Add bounded digital-PDF upload, extraction and page-linked source review, with manual correction for uncertain values. Complete one company case from upload through reviewed financial analysis.
Phase 4: Retrieval and AI Q&A	Add authorised retrieval, local-model question answering, citations and fallback text. Use reviewed values and tested Python calculations for numerical answers. Demonstrate supported answers and appropriate refusals.
Phase 5: Reporting, validation and delivery	Complete report generation, correction/versioning, deletion and audit events. Run end-to-end, formula and access control tests; freeze evaluation data; compare A/B/C; conduct an available usability review; and finalise demo instructions, limitations and the contribution log.

Teamwork and scope control

The three members will divide interface/usability, backend/data/security, extraction/financial logic, and retrieval/evaluation work, combining roles as needed. Each member should review another module and understand the full workflow. Keep a weekly decision log, issues, commits and test evidence.

If extraction takes longer, support fewer layouts and retain manual correction. If inference is slow, reduce model/context size. Cut optional packaging, charts or extra formats before cutting correct arithmetic, evidence checks or access control. Add features only when there is a specific user need, data source and test plan.

13. Expected Outcomes

The final output will be a local application, a tested six-indicator library, evidence/gap rules, an annotated data pack, a preparation report and reproducible evaluation results. Owners should be able to explain their financial concerns and identify the next information or investigation needed. We will test understanding and preparation effort, not promise loans, cheaper credit or higher revenue.

Proposed demonstration

Upload a fictional business's statements and correct one extracted value at its source. Show a margin change with its inputs. Leave interest expense missing: coverage must say "not assessed" and request that figure. Add the reviewed value, update the analysis and report, then show a denied cross-business access attempt and an unsupported loan-approval question.

Milestone 1 presents the proposal and planned work, not a completed product. The final demonstration must show live behaviour; cached results are only a labelled backup. [2]

References

Sources used in the full proposal; research access date: 24 October 2026. Product descriptions establish advertised capabilities, not independent performance results.

[1] Project team. CredLens: Financial Health & Financing Readiness Assistant. Agreed one-page brief, supplied attachment.

[2] Institution. Milestone 1: Project Proposal, Requirements & Feasibility, Section 7 and 30-mark rubric. Supplied sheet; deadline stated: 18 October 2026.

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